

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

ND

COURSE 09 OF 12
CURRICULUM GUIDE

Node.js for Teens & Beginners

Server-side JavaScript & backend APIs

GRADE BAND

Grade 9–12

DIFFICULTY

Beginner–Intermediate

DURATION

10 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

JavaScript beyond the browser. Students learn how servers actually work and build the backend APIs that power real applications, using Node.js and Express.

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Prerequisites: JavaScript for Kids & Teens.

LEARNING OUTCOMES

What Students Will Learn

- Understand how client-server communication works
- Build a server using Node.js and Express
- Design and implement RESTful API routes
- Read and write data using a simple database
- Handle requests, responses and basic error handling
- Connect a frontend to a self-built backend API

WHAT WE'LL USE

Tools & Technologies

Node.js Express npm Postman SQLite / JSON storage Git

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · How the Web Works	The client-server model and HTTP basics.
02 · Node.js Fundamentals	Running JavaScript on the server; modules.
03 · Building a Server with Express	Routes and middleware.

MODULE	FOCUS
04 · RESTful API Design	GET/POST/PUT/DELETE and status codes.
05 · Working with Data	Reading and writing to a database or file store.
06 · Testing APIs	Using Postman and handling errors.
07 · Connecting Frontend & Backend	The full request/response cycle.
08 · Capstone Backend Project	A complete, small API-powered application.

LEARNING BY BUILDING

Projects Students Will Build

- Notes / Task API — A full CRUD backend.
- Simple Blog or Bookmark API — Applied REST design.
- Capstone Full-Stack App — A frontend connected to a self-built API.

BEYOND THE SYNTAX

Skills Developed

Backend development REST API design Server-client architecture Debugging with Postman

PORTFOLIO OUTCOME

A deployed backend API with documentation and a working frontend demo.

RECOMMENDED NEXT STEP

SQL for Teens & Beginners, or DSA for Teens